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OCR GCSE Computer Science — **Network Protocols (J277/01 Section 1.3.2)**. This topic covers what protocols are, why they are needed, common protocols (TCP/IP, HTTP, HTTPS, FTP, POP, IMAP, SMTP), IP and MAC addressing, and the concept of layers in network communication.

Exam questions typically ask you to **describe** the purpose of specific protocols, **give examples** of valid IP and MAC addresses, **explain** why protocols are needed, and **identify** which protocol is appropriate for a given scenario. The 2024 exam included questions on IPv4/IPv6 address formats and MAC address structure.

What you must know

This checklist covers the key exam-visible concepts. You should be able to explain each point and give an example where appropriate.

- **Define** what a protocol is and explain why protocols are needed for network communication.
- **Describe** the purpose of common protocols: TCP/IP, HTTP, HTTPS, FTP, POP, IMAP, SMTP.
- **Give** valid examples of IPv4 addresses, IPv6 addresses, and MAC addresses.
- **Describe** the format and structure of IP addresses and MAC addresses.

- **Explain** the difference between IP addresses and MAC addresses.
- **Explain** the concept of layers in networking and why they are used.
- **Identify** which protocol is appropriate for different scenarios (web browsing, email, file transfer).

> **Specification note.** OCR J277 Section 1.3.2 requires understanding of modes of connection, encryption, IP/MAC addressing, standards, common protocols (TCP/IP, HTTP, HTTPS, FTP, POP, IMAP, SMTP), and the concept of layers. The 2024 exam asked for valid IPv4, IPv6, and MAC address examples.

Toolkit

What is a protocol?

A **protocol** is a set of rules that govern how data is transmitted between devices on a network. Protocols ensure that devices from different manufacturers can communicate with each other by defining a common "language" and format for data exchange. Without protocols, devices would not understand each other's messages.

Exam-ready definition: "A protocol is a set of rules that controls how data is transmitted across a network."

Why are protocols needed?

Reason	Explanation	Real-world analogy
Standardisation	Ensures all devices follow the same rules regardless of manufacturer	Like all countries agreeing to drive on the same side of the road
Compatibility	Allows different types of hardware and software to communicate	Like having a universal electrical plug standard
Error handling	Defines how to detect and correct errors in transmission	Like requiring a signature to confirm delivery
Data formatting	Specifies how data should be structured and packaged	Like addressing an envelope in a standard format
Security	Can include encryption and authentication requirements	Like sealing a letter in an envelope

Common protocols — quick reference

Protocol	Full name	Purpose	Port
TCP/IP	Transmission Control Protocol / Internet Protocol	Foundation of internet communication; breaks data into packets and routes them	Various
HTTP	Hypertext Transfer Protocol	Transfers web pages from servers to browsers (not secure)	80
HTTPS	HTTP Secure	Secure version of HTTP; encrypts data for safe transmission	443
FTP	File Transfer Protocol	Transfers files between computers on a network	21
SMTP	Simple Mail Transfer Protocol	Sends outgoing emails from client to server	25/587
POP	Post Office Protocol	Downloads emails from server to client (usually deletes from server)	110
IMAP	Internet Message Access Protocol	Accesses emails on server without downloading (syncs across devices)	143

IP addresses – format and examples

IPv4 (Internet Protocol version 4)

- 32-bit address
- Written as 4 decimal numbers separated by dots
- Each number ranges from 0 to 255
- Example: `192.168.1.1`
- Example: `86.134.23.199`
- About 4.3 billion unique addresses

IPv6 (Internet Protocol version 6)

- 128-bit address
- Written as 8 groups of 4 hexadecimal digits separated by colons
- Each group ranges from 0000 to FFFF
- Example: `2001:0db8:85a3:0000:0000:8a2e:0370:7334`
- Shortened: `2001:db8:85a3::8a2e:370:7334`
- Vastly more addresses than IPv4

MAC addresses – format and examples

A **MAC address** (Media Access Control address) is a unique identifier assigned to a network interface card (NIC) by the manufacturer. It is sometimes called a physical address or hardware address.

- 48-bit address (6 bytes)
- Written as 6 pairs of hexadecimal digits separated by colons or hyphens
- Each pair ranges from 00 to FF
- Example: `00:1A:2B:3C:4D:5E`
- Example: `A4-C3-F0-12-34-56`
- Permanently assigned at manufacture (usually cannot be changed)
- First 3 pairs identify the manufacturer, last 3 pairs identify the device

IP address vs MAC address

Feature	IP address	MAC address
What it identifies	A device's location on a network	The physical network interface card
Assigned by	Network administrator or DHCP server	Manufacturer (at production)
Can it change?	Yes — changes when connecting to different networks	No — permanently assigned (usually)
Format	Decimal (IPv4) or hexadecimal (IPv6)	Hexadecimal pairs
Used for	Routing data across networks (logical addressing)	Identifying devices on local network (physical addressing)
Analogy	Your postal address (can move house)	Your fingerprint (permanent and unique)

The concept of layers

Network communication is organised into **layers**, where each layer handles a specific part of the communication process. This is like an assembly line where each worker has one specific job.

Why use layers?

- **Simplification** — complex tasks are broken into manageable parts
- **Standardisation** — each layer has a defined interface with adjacent layers
- **Flexibility** — one layer can be updated without affecting others
- **Interoperability** — different manufacturers can create compatible products for each layer

Layer example: When you send an email, one layer handles the email format (SMTP), another handles breaking the message into packets (TCP), another handles addressing and routing (IP), and another handles the physical transmission (e.g., Ethernet).

1. What are protocols?

Imagine trying to have a conversation where one person speaks English and the other speaks French, with neither understanding the other's language. Communication would be impossible. **Protocols** solve this problem for computers by providing a common set of rules that all devices agree to follow.

The need for protocols

When computers communicate across a network, they face several challenges:

- Devices may be made by different manufacturers
- They may run different operating systems
- They may be connected by different types of cables or wireless signals
- Data may need to travel through many intermediate devices

Protocols provide **standardised rules** that overcome these differences. They define how data should be formatted, how errors should be handled, and how devices should respond to each other.

What protocols define

A protocol typically specifies:

- **Data format** — how information is structured
- **Addressing** — how to identify sender and receiver
- **Routing** — how data finds its path through the network
- **Error detection** — how to identify corrupted data
- **Error correction** — what to do when errors occur
- **Flow control** — how to manage the speed of transmission
- **Sequence** — the order of messages in a conversation

Everyday analogy

Sending data across a network is like sending a letter through the postal system:

- The **address format** must follow rules (postcode, street name, etc.)
- The **envelope** must be a standard size
- The **stamp** indicates payment has been made
- The postal service knows how to **route** the letter
- If the letter is lost, there may be **tracking** or confirmation

Network protocols work similarly — they define the "rules of the postal system" for digital data.

Exam tip — defining "protocol".
The most common exam question is "What is a protocol?" or "Define the term protocol."
Safe answer: "A protocol is a set of rules that controls how data is transmitted across a network."

For extra marks, add: "Protocols ensure that devices from different manufacturers can communicate by following standardised rules."

Key point — standardisation. The most important function of protocols is **standardisation**. Without agreed standards, a Windows computer couldn't communicate with a Mac, your phone couldn't connect to the router, and the internet simply wouldn't work. Protocols like TCP/IP are the reason billions of different devices can all communicate.

Worked Example 1 – Defining a protocol

Question: Define the term "protocol" in the context of computer networks. [2 marks]

Step 1: State that it is a set of rules.
Step 2: Explain what the rules govern.
Answer: A protocol is a set of rules [1 mark] that controls/governs how data is transmitted/formatted/exchanged across a network [1 mark].

Exam tip: Always include both "set of rules" AND what those rules are for. Just saying "rules for computers" is too vague.

Worked Example 2 – Explaining why protocols are needed

Question: Explain why protocols are needed for devices to communicate on a network. [3 marks]

Step 1: Identify the problem protocols solve.
Step 2: Explain how protocols solve it.
Answer:

- Devices on a network may be made by different manufacturers / run different operating systems [1 mark]
- Protocols provide standardised rules that all devices follow [1 mark]
- This allows devices to understand each other's messages / ensures compatibility / allows interoperability [1 mark]

Exam tip: Link your points together. Don't just list facts — show how they connect (problem → solution → benefit).

Worked Example 3 – Identifying protocol functions

Question: State two functions that a network protocol might define. [2 marks]

Choose any two from:

- How data is formatted / structured [1 mark]
- How errors are detected / corrected [1 mark]
- How devices are addressed / identified [1 mark]
- How data is routed through the network [1 mark]
- The speed / flow control of data transmission [1 mark]
- How data is encrypted / secured [1 mark]

Exam tip: Be specific. "How data is sent" is too vague. Say "how data is formatted" or "how errors are detected."

Worked Example 4 – Real-world analogy

Question: Using a real-world analogy, explain why standardised protocols are important for network communication. [3 marks]

Answer:
Protocols are like the rules of a postal system [1 mark]. Letters must have addresses written in a standard format (postcode, street, etc.) so that postal workers know where to deliver them [1 mark]. Similarly, network protocols define standard formats for data so that all devices can understand and correctly route the information [1 mark].
Alternative analogy: Like speaking the same language — without a common language (protocol), two people cannot communicate, even if they both have the ability to speak.

Exam tip: Analogy questions require you to make a clear link between the analogy and the computing concept. Don't just describe the analogy — explain what it represents.

2. Common protocols

Different protocols are designed for different purposes. Just as you might use a phone call for urgent matters and email for detailed information, different network protocols are suited to different types of data transfer. You need to know the purpose of each protocol and when it would be used.

TCP/IP – the foundation

TCP (Transmission Control Protocol)

- Breaks data into **packets**
- Ensures packets arrive **complete and in order**
- Requests **retransmission** of lost packets

- Reliable but slower due to error checking

IP (Internet Protocol)

- Handles **addressing** (IP addresses)
- **Routes** packets across networks
- Does not guarantee delivery
- Works with TCP to provide reliable communication

Together, TCP/IP forms the **foundation of the internet**. Almost all internet communication uses TCP/IP as its base protocol.

HTTP and HTTPS – web browsing

HTTP (Hypertext Transfer Protocol)

- Transfers **web pages** from servers to browsers
- Uses a **request-response** model
- Data is sent in **plain text** (not encrypted)
- Should NOT be used for sensitive data

HTTPS (HTTP Secure)

- Secure version of HTTP
- Data is **encrypted** before transmission
- Uses **SSL/TLS** certificates for security
- Essential for login pages, banking, shopping
- Shown by padlock icon in browser

FTP – file transfer

FTP (File Transfer Protocol)

- Transfers **files** between computers
- Can upload and download files
- Often used by web developers to upload website files
- Supports **authentication** (username/password)
- Not encrypted by default — secure alternatives such as SFTP or FTPS can be used

FTP is commonly used in scenarios where large files need to be transferred, such as uploading a website to a web server or downloading software updates.

Email protocols

SMTP (Simple Mail Transfer Protocol)

- **Sends** outgoing emails
- Pushes email from client to server
- Also transfers between mail servers

POP (Post Office Protocol)

- **Downloads** emails from server
- Usually **deletes** from server after download
- Emails stored locally on one device

IMAP (Internet Message Access Protocol)

- **Accesses** emails on server
- **Keeps** emails on server
- **Syncs** across multiple devices
- Better for accessing email from phone, tablet, and computer

Exam tip — POP vs IMAP.
A common exam question asks for the difference between POP and IMAP.
Key difference: POP downloads emails and usually deletes them from the server (one device access). IMAP keeps emails on the server and syncs across devices (multi-device access).
Scenario hint: If the question mentions "accessing email on multiple devices," the answer is IMAP.

Choosing the right protocol

Scenario	Protocol	Reason
Viewing a news website	HTTP or HTTPS	Requesting web pages from a server
Online banking	HTTPS	Sensitive data must be encrypted
Uploading files to a web server	FTP	Designed for file transfer
Sending an email	SMTP	Protocol for outgoing mail
Reading email on phone and laptop	IMAP	Syncs across multiple devices
Reading email on one device only	POP	Downloads to local device

Worked Example 5 – Describing HTTP and HTTPS

Question: Describe the purpose of HTTP and explain why HTTPS should be used instead for online banking. [4 marks]

HTTP purpose [2 marks]:

- HTTP is used to transfer web pages / hypertext documents [1 mark]
- from a web server to a web browser / client [1 mark]

Why HTTPS for banking [2 marks]:

- HTTPS encrypts the data being transmitted [1 mark]

- This prevents attackers from intercepting sensitive information such as passwords or account numbers / protects against man-in-the-middle attacks [1 mark]

Exam tip: For HTTPS questions, always mention encryption and give an example of what could be stolen (passwords, card details, personal data).

Worked Example 6 — Email protocols scenario

Question: Sarah wants to send an email and then read it on both her phone and laptop. State the protocol used to send the email and the protocol that would allow her to read it on multiple devices. Explain your choice. [4 marks]

Sending email [1 mark]: SMTP
Explanation [1 mark]: SMTP (Simple Mail Transfer Protocol) is used to send outgoing emails from a client to a mail server.
Reading on multiple devices [1 mark]: IMAP
Explanation [1 mark]: IMAP keeps emails on the server and synchronises them across multiple devices, so Sarah can see the same emails on her phone and laptop.

Exam tip: "Multiple devices" is a strong hint for IMAP. "Download and delete" hints at POP.

Worked Example 7 — FTP scenario

Question: A web developer has created a website on their local computer and needs to make it available on the internet. State an appropriate protocol for this task and explain why it is suitable. [3 marks]

Protocol [1 mark]: FTP (File Transfer Protocol)
Explanation [2 marks]:

- FTP is designed for transferring files between computers [1 mark]
- It allows the developer to upload all the website files (HTML, images, CSS) from their local computer to the web server [1 mark]

Exam tip: FTP questions often involve uploading website files, downloading large files, or transferring files between servers.

Worked Example 8 – TCP/IP explanation

Question: Describe the roles of TCP and IP in network communication. [4 marks]

TCP [2 marks]:

- TCP breaks data into packets / segments [1 mark]
- TCP ensures packets arrive complete and in the correct order / handles error checking and retransmission [1 mark]

IP [2 marks]:

- IP handles addressing — each device is identified by an IP address [1 mark]
- IP routes packets across the network to the correct destination [1 mark]

Exam tip: Remember: TCP = reliable delivery (packets, order, error checking). IP = addressing and routing.

3. Addressing and layers

For data to reach its destination, devices need addresses. Networks use two types of addresses: **IP addresses** for logical/network identification and **MAC addresses** for physical/hardware identification. Understanding the format of these addresses and how to give valid examples is essential for the exam.

IPv4 addresses in detail

IPv4 is the most common type of IP address you'll encounter. It consists of four numbers separated by dots, known as "dotted decimal" notation.

- Each number is called an **octet** (8 bits)
- Each octet ranges from **0 to 255**
- Total: 32 bits (4 × 8 bits)

Valid examples:

- `192.168.0.1` (common home router)
- `10.0.0.50` (private network)
- `172.16.254.1` (private network)
- `8.8.8.8` (Google's DNS server)

Invalid examples:

- `192.168.0.256` — 256 is out of range
- `192.168.0` — only 3 octets
- `192.168.0.1.5` — 5 octets

IPv6 addresses in detail

IPv6 was created because IPv4 addresses were running out. It uses a much larger address space.

- 8 groups of 4 hexadecimal digits
- Groups separated by **colons**
- Each digit is 0-9 or A-F
- Total: 128 bits

Valid examples:

- `2001:0db8:85a3:0000:0000:8a2e:0370:7334`
- `fe80:0000:0000:0000:0000:0000:0000:0001`

Shortened form: Leading zeros can be omitted, consecutive zero groups can be replaced with `::`

- `2001:db8:85a3::8a2e:370:7334`
- `fe80::1`

MAC addresses in detail

Every network interface card (NIC) has a unique MAC address assigned by the manufacturer. This is a **physical** or **hardware** address.

- 6 pairs of hexadecimal digits
- Separated by **colons** or **hyphens**
- Each pair ranges from **00 to FF**
- Total: 48 bits (6 × 8 bits)

Valid examples:

- `00:1A:2B:3C:4D:5E`
- `A4-C3-F0-12-34-56`
- `FF:FF:FF:FF:FF:FF` (broadcast)

Structure:

- First 3 pairs: **OUI** (Organizationally Unique Identifier) — identifies the manufacturer
- Last 3 pairs: **NIC specific** — unique to the device

> **Exam tip — address format questions (2024 exam).**
> The 2024 OCR exam asked students to:
> - Give a valid example of an IPv4 address
> - Give a valid example of an IPv6 address
> - Describe the format of a MAC address
>
> **Safe IPv4:** `192.168.1.1` — easy to remember, clearly valid.
> **Safe IPv6:** `2001:0db8:0000:0000:0000:0000:0000:0001` — all 8 groups shown.
> **Safe MAC:** `00:1A:2B:3C:4D:5E` — 6 pairs, hex digits, colons.

The concept of layers

Network communication is organised into **layers**, with each layer responsible for a specific aspect of the communication process. The most common model is the **TCP/IP model**, which has four layers. Each layer only communicates with the layers directly above and below it.

Layer	Name	Responsibility	Example protocols
4	Application	User-facing services; data format for applications	HTTP, HTTPS, FTP, SMTP, POP, IMAP
3	Transport	End-to-end communication; reliability; packet ordering	TCP, UDP
2	Internet	Addressing and routing across networks	IP

Layer	Name	Responsibility	Example protocols
1	Network Access / Link	Physical transmission; hardware addressing	Ethernet, Wi-Fi

Benefits of layered architecture

- **Modularity** — each layer can be developed independently
- **Flexibility** — one layer can be updated without affecting others
- **Interoperability** — different manufacturers can create compatible products
- **Troubleshooting** — problems can be isolated to specific layers
- **Abstraction** — higher layers don't need to know the details of lower layers

Exam tip — layers questions.
You don't need to memorise all layer names for OCR GCSE, but you should understand:

- Communication is divided into layers
- Each layer has a specific job
- Layers make networks easier to design, update, and troubleshoot

Worked Example 9 — Valid IP address examples [2024 exam style]

Question: Give one valid example of an IPv4 address and one valid example of an IPv6 address. [2 marks]

IPv4 [1 mark]: `192.168.1.1`
Validation: Four numbers separated by dots, each number between 0-255. ✓
IPv6 [1 mark]: `2001:0db8:85a3:0000:0000:8a2e:0370:7334`
Validation: Eight groups of four hexadecimal digits separated by colons. ✓
Other acceptable IPv4 answers: 10.0.0.1, 172.16.0.1, 8.8.8.8, 255.255.255.0
Other acceptable IPv6 answers: fe80::1, 2001:db8::1 (shortened forms are valid)

Exam tip: Use memorable addresses. 192.168.1.1 is a common router address. Don't use numbers above 255 for IPv4 or letters above F for IPv6.

Worked Example 10 — MAC address format [2024 exam style]

Question: Describe the format of a MAC address. [2 marks]

Answer:

- A MAC address consists of 6 pairs / 12 digits / 48 bits [1 mark]
- of hexadecimal characters / numbers and letters A-F [1 mark]
- separated by colons or hyphens [can gain this mark if shown in example]

Example to support answer: `00:1A:2B:3C:4D:5E`

Exam tip: If asked to "describe the format," include: number of parts (6 pairs), the number system (hexadecimal), and the separator (colons or hyphens).

Worked Example 11 — IP vs MAC address comparison

Question: Explain two differences between an IP address and a MAC address. [4 marks]

Difference 1 [2 marks]:
An IP address is assigned by the network / can change when connecting to different networks [1 mark], whereas a MAC address is assigned by the manufacturer and is permanent / cannot usually be changed [1 mark].
Difference 2 [2 marks]:
An IP address identifies a device's logical location on a network for routing purposes [1 mark], whereas a MAC address identifies the physical hardware / network interface card [1 mark].
Alternative differences:

- IP addresses are used for routing between networks; MAC addresses are used within a local network
- IPv4 uses decimal notation (e.g., 192.168.1.1); MAC uses hexadecimal (e.g., 00:1A:2B:3C:4D:5E)

Exam tip: For comparison questions, structure your answer clearly: "X is... whereas Y is..." This makes it obvious you're giving both sides.

Worked Example 12 — Explaining the benefit of layers

Question: A network system uses a layered architecture for communication. Explain two benefits of using layers. [4 marks]

Benefit 1 [2 marks]:
Each layer can be developed or updated independently [1 mark], so changes to one layer do not require changes to other layers / makes maintenance easier [1 mark].
Benefit 2 [2 marks]:
Different manufacturers can create products for specific layers that work together [1 mark], as long as they follow the protocol standards for that layer / promotes interoperability [1 mark].
Alternative benefits:

- Problems can be isolated to a specific layer, making troubleshooting easier

- Complex communication is broken into smaller, manageable tasks
- Abstraction means higher layers don't need to understand the technical details of lower layers

Exam tip: Always give the benefit AND its consequence. "Can be updated independently" alone might be 1 mark. Adding "so changes don't affect other layers" gains the second mark.

Summary of worked examples

#	Topic	Question type	Key skill
1	Protocol definition	Define term	"Set of rules" + what they govern
2	Why protocols needed	Explain	Problem → solution → benefit chain
3	Protocol functions	State functions	Specific functions (formatting, error handling)
4	Protocol analogy	Real-world link	Analogy + computing connection
5	HTTP vs HTTPS	Describe + explain	Encryption for sensitive data
6	Email protocols	Scenario application	SMTP sends, IMAP syncs
7	FTP usage	Scenario application	File transfer to web server
8	TCP/IP roles	Describe roles	TCP = reliable, IP = addressing
9	IP address examples	Give valid examples (2024)	Correct format for IPv4/IPv6
10	MAC address format	Describe format (2024)	6 pairs, hexadecimal, colons/hyphens
11	IP vs MAC	Compare/contrast	Logical vs physical, changeable vs permanent
12	Layers benefits	Explain benefits	Independence, interoperability

Quick reference

Key terms

Term	Definition (exam-answer style)
Protocol	A set of rules that controls how data is transmitted across a network

Term	Definition (exam-answer style)
TCP/IP	The foundational protocol suite for the internet; TCP handles reliable transmission, IP handles addressing and routing
HTTP	Protocol for transferring web pages from servers to browsers; not encrypted
HTTPS	Secure version of HTTP; encrypts data for safe transmission of sensitive information
FTP	Protocol for transferring files between computers on a network
SMTP	Protocol for sending outgoing emails from client to mail server
POP	Protocol for downloading emails from server; usually deletes from server after download
IMAP	Protocol for accessing emails on server; keeps emails on server and syncs across devices
IP address	A logical address assigned to a device to identify it on a network; can change
MAC address	A physical/hardware address permanently assigned to a network interface card by the manufacturer
IPv4	32-bit IP address format: four decimal numbers (0-255) separated by dots
IPv6	128-bit IP address format: eight groups of four hexadecimal digits separated by colons
Layers	A way of organising network communication where each layer handles a specific task
Encryption	Converting data into a coded form to prevent unauthorised access

Protocol purposes – quick lookup

Protocol	One-line purpose	Memory aid
HTTP	Web pages (not secure)	**H**ypertext = web
HTTPS	Secure web pages	**S** = Secure
FTP	File transfer	**F**ile **T**ransfer
SMTP	Send email	**S**end **M**ail
POP	Download email (one device)	**P**ost **O**ffice = pickup and take away

Protocol	One-line purpose	Memory aid
IMAP	Access email (multiple devices)	**I** **M**anage **A**cross **P**hones

Address formats at a glance

Address type	Format	Example	Key fea
IPv4	4 numbers (0-255), dots	192.168.1.1	32 bits, decima
IPv6	8 groups of 4 hex digits, colons	2001:0db8:85a3:0000:0000:8a2e:0370:7334	128 bits hexade
MAC	6 pairs of hex digits, colons/hyphens	00:1A:2B:3C:4D:5E	48 bits, permar

Typical pitfalls

Pitfall 1 — Vague protocol definition.
Wrong: "A protocol is how computers talk to each other."
Right: "A protocol is a **set of rules** that controls how data is **transmitted across a network**."

Pitfall 2 — IPv4 numbers out of range.
Wrong: `192.168.1.256` (256 > 255)
Right: `192.168.1.255` or lower. Each octet must be **0-255**.

Pitfall 3 — Wrong number of IPv4 octets.
Wrong: `192.168.1` (3 octets) or `192.168.1.1.5` (5 octets)
Right: IPv4 always has exactly **4 numbers** separated by **3 dots**.

Pitfall 4 — Confusing HTTP and HTTPS usage.
Wrong: "HTTP is fine for online banking."
Right: HTTPS must be used for **sensitive data** (banking, login, payments) because it **encrypts** the data.

Pitfall 5 — Confusing POP and IMAP.
Wrong: "POP syncs emails across devices."
Right: **IMAP** syncs across devices. **POP** downloads to one device (and usually deletes from server).

Pitfall 6 — Wrong characters in MAC address.
Wrong: `00:1G:2B:3C:4D:5E` (G is not valid hex)
Right: MAC addresses use hexadecimal: **0-9 and A-F only**.

Pitfall 7 — Thinking MAC addresses can change.
Wrong: "MAC addresses change when you connect to a new network."
Right: MAC addresses are **permanently assigned by the manufacturer**. IP addresses can change.

Pitfall 8 — Using dots in IPv6.
Wrong: `2001.0db8.85a3.0000.0000.8a2e.0370.7334`
Right: IPv6 uses **colons**, not dots: `2001:0db8:85a3:...`

Pitfall 9 — Mixing up sending and receiving email protocols.
Wrong: "SMTP is used to receive emails."
Right: SMTP sends emails. **POP and IMAP receive** emails.

Pitfall 10 — Not explaining WHY layers are beneficial.
Wrong: "Layers divide communication into parts."
Right: "Layers allow each part to be **developed independently**, so one layer can be **updated without affecting others**."

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